CLOCK_50
RESET_N
UART_RX

uart_rx:RX0

clk

resetrx data_valid

data_out[7..0]

a_reg~[7..0] 01

n_reg~[7..0] 01

a_reg~[15..8] 01

n_reg~[15..8] 01

calc_start D
CLKCLRN

SCLR

1'h0 Q

a_reg[7..0] D
CLKCLRN

SCLR

8'h0 Q

n_reg[7..0] D
CLKCLRN

SCLR

8'h0 Q =Equal0
A[31..0] 32'hc B[31..0] OUT +

1'h0 CIN Add0

A[3..0] 4'h1 B[3..0] OUT[3..0]

potencia_q8020:POWER0

clk

reset start

a[7..0] n[7..0] done

result_q[99..0]

byte_index~[7..4] 01

WideOr0

result_reg~[99..0] 01

result_reg~[199..100] 01

byte_index~[3..0] 0 4'h0 1

byte_index~[11..8] 01

Selector38

SEL[2..0] DATA[2..0] OUT Selector39
SEL[2..0] DATA[2..0] OUT Selector40
SEL[2..0] DATA[2..0] OUT Selector41
SEL[2..0] DATA[2..0] OUT Selector42
SEL[2..0] DATA[2..0] OUT Selector66
SEL[2..0] DATA[2..0] OUT Selector67
SEL[2..0] DATA[2..0] OUT Selector68
SEL[2..0] DATA[2..0] OUT Selector69
SEL[2..0] DATA[2..0] OUT Selector70
SEL[2..0] DATA[2..0] OUT Selector71
SEL[2..0] DATA[2..0] OUT Selector72
SEL[2..0] DATA[2..0] OUT Selector73
SEL[2..0] DATA[2..0] OUT Selector74
SEL[2..0] DATA[2..0] OUT Selector75
SEL[2..0] DATA[2..0] OUT Selector76
SEL[2..0] DATA[2..0] OUT Selector77
SEL[2..0] DATA[2..0] OUT Selector78
SEL[2..0] DATA[2..0] OUT Selector79
SEL[2..0] DATA[2..0] OUT Selector80
SEL[2..0] DATA[2..0] OUT Selector81
SEL[2..0] DATA[2..0] OUT Selector82
SEL[2..0] DATA[2..0] OUT Selector83
SEL[2..0] DATA[2..0] OUT Selector84
SEL[2..0] DATA[2..0] OUT Selector85
SEL[2..0] DATA[2..0] OUT Selector86
SEL[2..0] DATA[2..0] OUT Selector87
SEL[2..0] DATA[2..0] OUT Selector88
SEL[2..0] DATA[2..0] OUT Selector89
SEL[2..0] DATA[2..0] OUT Selector90
SEL[2..0] DATA[2..0] OUT Selector91
SEL[2..0] DATA[2..0] OUT Selector92
SEL[2..0] DATA[2..0] OUT Selector93
SEL[2..0] DATA[2..0] OUT Selector94
SEL[2..0] DATA[2..0] OUT Selector95
SEL[2..0] DATA[2..0] OUT Selector96
SEL[2..0] DATA[2..0] OUT Selector97
SEL[2..0] DATA[2..0] OUT Selector7
SEL[2..0] DATA[2..0] OUT Selector8
SEL[2..0] DATA[2..0] OUT Selector9
SEL[2..0] DATA[2..0] OUT Selector10
SEL[2..0] DATA[2..0] OUT Selector11
SEL[2..0] DATA[2..0] OUT Selector12
SEL[2..0] DATA[2..0] OUT Selector13
SEL[2..0] DATA[2..0] OUT Selector14
SEL[2..0] DATA[2..0] OUT Selector15
SEL[2..0] DATA[2..0] OUT Selector16
SEL[2..0] DATA[2..0] OUT Selector17
SEL[2..0] DATA[2..0] OUT Selector18
SEL[2..0] DATA[2..0] OUT Selector19
SEL[2..0] DATA[2..0] OUT Selector20
SEL[2..0] DATA[2..0] OUT Selector21
SEL[2..0] DATA[2..0] OUT Selector22
SEL[2..0] DATA[2..0] OUT Selector23
SEL[2..0] DATA[2..0] OUT Selector24
SEL[2..0] DATA[2..0] OUT Selector25
SEL[2..0] DATA[2..0] OUT Selector26
SEL[2..0] DATA[2..0] OUT Selector27
SEL[2..0] DATA[2..0] OUT Selector28
SEL[2..0] DATA[2..0] OUT Selector29
SEL[2..0] DATA[2..0] OUT Selector30
SEL[2..0] DATA[2..0] OUT Selector31
SEL[2..0] DATA[2..0] OUT Selector32
SEL[2..0] DATA[2..0] OUT Selector33
SEL[2..0] DATA[2..0] OUT Selector34
SEL[2..0] DATA[2..0] OUT Selector35
SEL[2..0] DATA[2..0] OUT Selector36
SEL[2..0] DATA[2..0] OUT Selector37
SEL[2..0] DATA[2..0] OUT Selector43
SEL[2..0] DATA[2..0] OUT Selector44
SEL[2..0] DATA[2..0] OUT Selector45
SEL[2..0] DATA[2..0] OUT Selector46
SEL[2..0] DATA[2..0] OUT Selector100
SEL[2..0] DATA[2..0] OUT Selector101
SEL[2..0] DATA[2..0] OUT Selector102
SEL[2..0] DATA[2..0] OUT Selector103
SEL[2..0] DATA[2..0] OUT Selector104
SEL[2..0] DATA[2..0] OUT Selector105
SEL[2..0] DATA[2..0] OUT Selector106
SEL[2..0] DATA[2..0] OUT Selector47
SEL[2..0] DATA[2..0] OUT Selector48
SEL[2..0] DATA[2..0] OUT Selector49
SEL[2..0] DATA[2..0] OUT Selector50
SEL[2..0] DATA[2..0] OUT Selector51
SEL[2..0] DATA[2..0] OUT Selector52
SEL[2..0] DATA[2..0] OUT Selector53
SEL[2..0] DATA[2..0] OUT Selector54
SEL[2..0] DATA[2..0] OUT Selector55
SEL[2..0] DATA[2..0] OUT Selector56
SEL[2..0] DATA[2..0] OUT Selector57
SEL[2..0] DATA[2..0] OUT Selector58
SEL[2..0] DATA[2..0] OUT Selector59
SEL[2..0] DATA[2..0] OUT Selector60
SEL[2..0] DATA[2..0] OUT Selector61
SEL[2..0] DATA[2..0] OUT Selector62
SEL[2..0] DATA[2..0] OUT Selector63
SEL[2..0] DATA[2..0] OUT Selector64
SEL[2..0] DATA[2..0] OUT Selector65
SEL[2..0] DATA[2..0] OUT Selector98
SEL[2..0] DATA[2..0] OUT Selector99
SEL[2..0] DATA[2..0] OUT

Selector107

SEL[2..0] DATA[2..0] OUT Selector110

SEL[2..0] DATA[2..0] OUT Selector109
SEL[2..0] DATA[2..0] OUT Selector108
SEL[2..0] DATA[2..0] OUT

state~35 1'h0

result_reg[99..0] D
CLKCLRN

SCLR

100'h0 Q

byte_index[3..0] D
CLKCLRN

SCLR

4'h0 Q

state

clk potencia_q8020:POWER0
reset uart_rx:RX0
uart_tx:TX0

byte_index[3..0] CALC

SEND WAIT_A WAIT_BUSY WAIT_CALC WAIT_IDLE WAIT_N
tx_data~[7..0] 01

tx_start 1'h0 01

tx_data~[15..8] 01

tx_start D
CLKCLRN

SCLR

1'h0 Q

tx_data[7..0] D
CLKCLRN

SCLR

8'h0 Q

uart_tx:TX0

clk
reset
tx_start

data_in[7..0] busy

tx UART_TX 8:99 68->DATA[2]16505 DATA[2]16513<-67 DATA[2]16521<-66 DATA[2]16529<-65 DATA[2]16537<-64 DATA[2]16729<-40 DATA[2]16737<-39 DATA[2]16745<-38 DATA[2]16753<-37 DATA[2]16761<-36 DATA[2]16769<-35 DATA[2]16777<-34 DATA[2]16785<-33 DATA[2]16793<-32 DATA[2]16801<-31 DATA[2]16809<-30 DATA[2]16817<-29 DATA[2]16825<-28 DATA[2]16833<-27 DATA[2]16841<-26 DATA[2]16849<-25 DATA[2]16857<-24 DATA[2]16865<-23 DATA[2]16873<-22 DATA[2]16881<-21 DATA[2]16889<-20 DATA[2]16897<-19 DATA[2]16905<-18 DATA[2]16913<-17 DATA[2]16921<-16 DATA[2]16929<-15 DATA[2]16937<-14 DATA[2]16945<-13 DATA[2]16953<-12 DATA[2]16961<-11 DATA[2]16969<-10 DATA[2]16977<-9 DATA[2]16257<-99 DATA[2]16265<-98 DATA[2]16273<-97 DATA[2]16281<-96 DATA[2]16289<-95 DATA[2]16297<-94 DATA[2]16305<-93 DATA[2]16313<-92 DATA[2]16321<-91 DATA[2]16329<-90 DATA[2]16337<-89 DATA[2]16345<-88 DATA[2]16353<-87 DATA[2]16361<-86 DATA[2]16369<-85 DATA[2]16377<-84 DATA[2]16385<-83 DATA[2]16393<-82 DATA[2]16401<-81 DATA[2]16409<-80 DATA[2]16417<-79 DATA[2]16425<-78 DATA[2]16433<-77 DATA[2]16441<-76 DATA[2]16449<-75 DATA[2]16457<-74 DATA[2]16465<-73 DATA[2]16473<-72 DATA[2]16481<-71 DATA[2]16489<-70 DATA[2]16497<-69 DATA[2]16545<-63 DATA[2]16553<-62 DATA[2]16561<-61 DATA[2]16569<-60 DATA[2]17001<-6 DATA[2]17009<-5 DATA[2]17017<-4 DATA[2]17025<-3 DATA[2]17033<-2 DATA[2]17041<-1 DATA[2]17049<-0 DATA[2]16577<-59 DATA[2]16585<-58 DATA[2]16593<-57 DATA[2]16601<-56 DATA[2]16609<-55 DATA[2]16617<-54 DATA[2]16625<-53 DATA[2]16633<-52 DATA[2]16641<-51 DATA[2]16649<-50 DATA[2]16657<-49 DATA[2]16665<-48 DATA[2]16673<-47 DATA[2]16681<-46 DATA[2]16689<-45 DATA[2]16697<-44 DATA[2]16705<-43 DATA[2]16713<-42 DATA[2]16721<-41 DATA[2]16985<-8 DATA[2]16993<-7

0:7 3:0 3->DATA[2]17057 DATA[2]17065<-2 DATA[2]17081<-0 DATA[2]17073<-1 DATA[1]17064<-1 0->DATA[1]17056 DATA[1]17080<-3 DATA[1]17072<-2 8->DATA[0]17055 DATA[0]17063<-9 DATA[0]17079<-11 DATA[0]17071<-10 31->DATA[1]16504 DATA[1]16992<-92 DATA[1]16984<-91DATA[1]16720<-58DATA[1]16712<-57DATA[1]16704<-56DATA[1]16696<-55DATA[1]16688<-54DATA[1]16680<-53DATA[1]16672<-52DATA[1]16664<-51DATA[1]16656<-50DATA[1]16648<-49DATA[1]16640<-48DATA[1]16632<-47DATA[1]16624<-46DATA[1]16616<-45DATA[1]16608<-44DATA[1]16600<-43DATA[1]16592<-42DATA[1]16584<-41DATA[1]16576<-40DATA[1]17048<-99DATA[1]17040<-98DATA[1]17032<-97DATA[1]17024<-96DATA[1]17016<-95DATA[1]17008<-94DATA[1]17000<-93DATA[1]16568<-39DATA[1]16560<-38DATA[1]16552<-37DATA[1]16544<-36DATA[1]16496<-30DATA[1]16488<-29DATA[1]16480<-28DATA[1]16472<-27DATA[1]16464<-26DATA[1]16456<-25DATA[1]16448<-24DATA[1]16440<-23DATA[1]16432<-22DATA[1]16424<-21DATA[1]16416<-20DATA[1]16408<-19DATA[1]16400<-18DATA[1]16392<-17DATA[1]16384<-16DATA[1]16376<-15DATA[1]16368<-14DATA[1]16360<-13DATA[1]16352<-12DATA[1]16344<-11DATA[1]16336<-10DATA[1]16328<-9DATA[1]16320<-8DATA[1]16312<-7DATA[1]16304<-6DATA[1]16296<-5DATA[1]16288<-4DATA[1]16280<-3DATA[1]16272<-2DATA[1]16264<-1DATA[1]16256<-0DATA[1]16976<-90DATA[1]16968<-89DATA[1]16960<-88DATA[1]16952<-87DATA[1]16944<-86DATA[1]16936<-85DATA[1]16928<-84DATA[1]16920<-83DATA[1]16912<-82DATA[1]16904<-81DATA[1]16896<-80DATA[1]16888<-79DATA[1]16880<-78DATA[1]16872<-77DATA[1]16864<-76DATA[1]16856<-75DATA[1]16848<-74DATA[1]16840<-73DATA[1]16832<-72DATA[1]16824<-71DATA[1]16816<-70DATA[1]16808<-69DATA[1]16800<-68DATA[1]16792<-67DATA[1]16784<-66DATA[1]16776<-65DATA[1]16768<-64DATA[1]16760<-63DATA[1]16752<-62DATA[1]16744<-61DATA[1]16736<-60DATA[1]16728<-59DATA[1]16536<-35DATA[1]16528<-34DATA[1]16520<-33DATA[1]16512<-32 131->DATA[0]16503 DATA[0]16991<-192 DATA[0]16983<-191DATA[0]16719<-158DATA[0]16711<-157DATA[0]16703<-156DATA[0]16695<-155DATA[0]16687<-154DATA[0]16679<-153DATA[0]16671<-152DATA[0]16663<-151DATA[0]16655<-150DATA[0]16647<-149DATA[0]16639<-148DATA[0]16631<-147DATA[0]16623<-146DATA[0]16615<-145DATA[0]16607<-144DATA[0]16599<-143DATA[0]16591<-142DATA[0]16583<-141DATA[0]16575<-140DATA[0]17047<-199DATA[0]17039<-198DATA[0]17031<-197DATA[0]17023<-196DATA[0]17015<-195DATA[0]17007<-194DATA[0]16999<-193DATA[0]16567<-139DATA[0]16559<-138DATA[0]16551<-137DATA[0]16543<-136DATA[0]16495<-130DATA[0]16487<-129DATA[0]16479<-128DATA[0]16471<-127DATA[0]16463<-126DATA[0]16455<-125DATA[0]16447<-124DATA[0]16439<-123DATA[0]16431<-122DATA[0]16423<-121DATA[0]16415<-120DATA[0]16407<-119DATA[0]16399<-118DATA[0]16391<-117DATA[0]16383<-116DATA[0]16375<-115DATA[0]16367<-114DATA[0]16359<-113DATA[0]16351<-112DATA[0]16343<-111DATA[0]16335<-110DATA[0]16327<-109DATA[0]16319<-108DATA[0]16311<-107DATA[0]16303<-106DATA[0]16295<-105DATA[0]16287<-104DATA[0]16279<-103DATA[0]16271<-102DATA[0]16263<-101DATA[0]16255<-100DATA[0]16975<-190DATA[0]16967<-189DATA[0]16959<-188DATA[0]16951<-187DATA[0]16943<-186DATA[0]16935<-185DATA[0]16927<-184DATA[0]16919<-183DATA[0]16911<-182DATA[0]16903<-181DATA[0]16895<-180DATA[0]16887<-179DATA[0]16879<-178DATA[0]16871<-177DATA[0]16863<-176DATA[0]16855<-175DATA[0]16847<-174DATA[0]16839<-173DATA[0]16831<-172DATA[0]16823<-171DATA[0]16815<-170DATA[0]16807<-169DATA[0]16799<-168DATA[0]16791<-167DATA[0]16783<-166DATA[0]16775<-165DATA[0]16767<-164DATA[0]16759<-163DATA[0]16751<-162DATA[0]16743<-161DATA[0]16735<-160DATA[0]16727<-159DATA[0]16535<-135DATA[0]16527<-134DATA[0]16519<-133DATA[0]16511<-132